MILITARY SPECIFICATION

RECEIVER, LOW FREQUENCY, MEDIUM FREQUENCY, AND HIGH FREQUENCY COMMUNICATIONS

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the operational characteristics, test, and acceptance requirements for a low frequency (LF), medium frequency (MF), and high frequency (HF) Communications Receiver, hereinafter referred to as the receiver.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-C-915	Cable And Cord Electrical, For Shipboard Use, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of
MIL-E-21981	Electronics Type Designations, Identification Plates And Markings: Requirements For
MIL-C-24056	Cabinets, Electronic Equipment, CY-4516(A)/S And CY-4516(B)/S (Single Bay) (Naval Shipboard Use)
MIL-P-28809	Printed Wiring Assemblies
MIL-C-39012	Connectors, Coaxial, Radio Frequency; General Specification For
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188	Military Communications System Technical Standards
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Electronic Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels And Associated Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70-1984	National Electrical Code

(Application for copies should be addressed to National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The receiver shall be in accordance with MIL-E-16400 to the extent specified herein. The receiver shall be a digitally tuned superheterodyne receiver with a frequency range of 14.0 kilohertz (kHz) to 30.0 megahertz (MHz). The receiver shall provide reception of single sideband (SSB), upper sideband (USB), lower sideband (LSB), and independent sideband (ISB), frequency modulation (FM), continuous wave (CW), amplitude modulation (AM) and, with an external converter, frequency-shift keyed (FSK) multi-channel radio teletype (see 6.4.2.2).

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Parts and materials. Parts and materials shall be as specified in 3.1.2.1 through 3.1.2.4.

3.1.2.1 Unacceptable materials. Receiver design shall not include the materials specified in a through f:

a. Polychlorinated biphenyls
b. Asbestos and asbestos compounds
c. Fragile or brittle materials
d. Beryllium and beryllium compounds, unless so identified
e. Lithium and lithium compounds unless specifically approved by the procuring activity
f. Flammable or combustible materials

The Unacceptable materials paragraph of MIL-E-16400 shall apply.

3.1.2.2 Individual selection. The performance of the receiver shall not be dependent on the selection of individual parts unless such selection is specifically approved by the procuring activity.

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3.1.2.3 Mercury and radioactive material. The Mercury and radioactive material paragraph of MIL-E-16400 shall apply. In addition, no parts shall have come in direct contact with metallic mercury, mercury compounds, or mercury bearing instruments or devices employing only a single boundary of containment during manufacture or inspection. (A single boundary of containment is one which is not backed up by a second seal or barrier to prevent escape and consequent mercury contamination in the event of rupture of the primary seal or barrier.)

3.1.2.4 Type BNC and Type N radio frequency (RF) connectors. Where Type BNC and Type N RF coaxial connectors are specified herein, connectors shall be selected in accordance with MIL-C-39012.

3.2 Operational requirements. Each receiver shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

3.3 Principal operating characteristics. Principal operating characteristics shall be as specified in 3.3.1 through 3.3.4.

3.3.1 Frequency range. The frequency (tuning) range of the receiver shall be from 14.0 kHz to 30.0 MHz. Each receiver shall be capable of adjustment to any frequency from 14.0 kHz to 29.99999 MHz.

3.3.2 Tuning accuracy. The receiver shall tune to the desired operating frequency with an accuracy equivalent to that of the frequency standard.

3.3.3 Method of tuning. Frequency selection over the complete frequency range of the receiver shall be accomplished in increments of 10 hertz (Hz) or less by front panel key pad entry or by a knob which shall provide manual tuning. A means shall be provided to rapidly increment or decrement the frequency in 10-Hz, 100-Hz, and 1-kHz steps.

3.3.4 Frequency display. The selected frequency shall be observable on a front panel numerical display device. The display shall be readable with the receiver located in a totally dark location. The display shall be readable with no difficulty at a viewing angle of 40 degrees with respect to the surface of the display

3.4 Primary power requirements. The receiver shall be designed to operate from 115 volts alternating current (VAC), 60-Hz, single-phase electrical power having characteristics as specified for Type I power in DoD-STD-1399, Section 300. The total power consumption shall not exceed 90 watts. A switch for disconnecting the receiver from primary power shall be mounted on the receiver front panel and the function clearly labeled. The switch shall break all power conductors of the power circuit. A green power indicator lamp shall be mounted near the switch to indicate when the receiver is energized.

3.4.1 Transient protection. The receiver shall operate without damage or permanent change in characteristics due to transient voltage, transient frequency, or power interruption as specified in MIL-E-16400 and the spike tests of DoD-STD-1399, Section 300, using a 2500-volt (V) spike for 115 V.

3.4.2 Overcurrent protection. Electrical overcurrent protection shall conform to the Circuit breakers, Fuses and fuse holders, and Electrical overload protection paragraphs of MIL-E-16400. When fuses are used, front panel blown fuse indicators shall be provided. When circuit breakers are used, a visible status indication shall be provided. A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded power conductor as determined by the National Electrical Code (NFPA 70-1984) for grounded source power. The overcurrent device shall be connected to the load side of the power switch.

3.4.3 Leakage current. The receiver leakage current measured in accordance with the procedure specified in 4.5.5.26 shall not exceed 5 milliamperes (mA).

3.5 Modes of operation. The receiver shall be capable of receiving by selection of one of the modes of operation specified in a through g:

a. LSB
b. USB
c. ISB
d. AM
e. CW telegraphy
f. FM
g. FSK

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3.6 Composition. The equipment shall consist of one each of the items specified in a and b:

a. Receiver
b. Set of mating connectors and cable clamps

3.6.1 Accessories. In addition to the basic equipment specified in 3.6, the accessory items specified in a and b shall be furnished in the quantities specified in the contract:

a. Mounting brackets for CY-4516, or equivalent, hard-mounted installation
b. Mounting brackets for EIA RS-310-C-77 mounting rack

3.7 Radio receiver system. The contractor shall be responsible for demonstrating the full range of operations of the receiver to be delivered in accordance with the requirements of this specification. Receiver provisions for external cabling to the radio receiver system shall be required. Standard Navy cables shall be as specified in MIL-C-915. Mating connectors for all external cabling which requires manufacture by the installing activity shall be furnished with each receiver. Receiver design shall preclude damage to electrical cabling during all conditions of assembly, removal, insertion, and tilting of the receiver.

3.8 Dimensions and weight. The maximum dimensions and weight of the receiver shall not exceed those shown in TABLE I.

TABLE I. Dimensions and weight.

Unit	Width cm 1/ (in.) 2/	Depth 3/ cm (in.)	Height cm (in.)	Weight kg 4/ (lb) 5/
Receiver	48.3 (19.0)	49.5 (19.5)	17.8 (7.0)	23 (50.6)

1/ Centimeters
2/ Inches
3/ Exclusive of rear connectors
4/ Kilograms
5/ Pounds

3.9 Environmental conditions. The receiver shall be capable of continuous, reliable operation, within the performance limits specified in 3.13 and the Environmental service condition paragraph specified in MIL-E-16400 and as specified in a through e:

a. Temperature Range 4
b. Relative humidity
c. Salt fog (spray) for sheltered equipment
d. Shock
e. Vibration

The receiver shall operate within the specified tolerances throughout the full range of environmental conditions without alignment or adjustment, other than external controls normally employed for operation of the receiver.

3.10 Orientation. The receiver shall be capable of storage in any orientation.

3.11 Basic performance conditions. The basic test conditions specified in 4.5 shall apply to the performance requirements specified in 3.13 except as modified for a specific test or test procedure specified in this specification.

3.12 Mechanical design and construction. Mechanical design and construction shall conform to the Mechanical design and construction paragraph of MIL-E-16400 except as modified herein.

3.12.1 General requirements. General requirements shall be as specified in 3.12.1.1 through 3.12.1.28.

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3.12.1.1 Construction. Construction shall conform to the Construction paragraph of MIL-E-16400. The receiver shall consist basically of a chassis, a front panel, and a group of replaceable plug-in assemblies. The chassis shall include the power supply, receptacles for the insertion of the plug-in assemblies, RF circuit protection, and the mechanical hardware associated with frequency selection. The front panel shall contain the required control functions and be securely mounted to the chassis. Two handles shall be provided, one mounted on each side of the front panel and each capable of supporting the entire weight of the unit. There shall be no sharp corners or edges on cases, panels, or brackets. Each printed circuit assembly shall be keyed to prevent incorrect insertion.

3.12.1.2 Rack mounting. The chassis of the receiver shall be designed and constructed so that when mounting brackets are installed on the sides of the chassis and the receiver is installed in a rack and subjected to shock or vibration, no other support or reinforcing members will be required to prevent damage to any part of the receiver.

3.12.1.2.1 Cabinet mounting. The receiver shall also be designed to permit installation in an Electronic Equipment Cabinet, CY-4516(A)/U as specified in MIL-C-24056.

3.12.1.3 Bonding and grounding. The receiver shall conform to the Bonding and grounding paragraph of MIL-E-16400. Additionally, a stud or pad grounding facility shall be provided. The facility shall be a pressed-in captive insert integral with the chassis rear panel and constructed of austenitic corrosion-resistant steel. The insert shall be flush with the outside surface of the chassis. If a pad is used, the pad shall contain a blind threaded hole to accommodate a 5/16 in. - 18 bolt for ground strap fastening. If a stud is used, the stud shall be 5/16 in. - 18 threaded. A 25.4-millimeter (mm) (1 in.) diameter area centered on the blind hole or stud shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.12.1.4 Corrosion protection, corrosion-resisting treatment, and painting. The receiver shall conform to the Corrosion protection/corrosion-resisting treatments paragraph and the Painting paragraph of MIL-E-16400.

3.12.1.5 Fastener hardware. The receiver shall conform to the Fastener hardware paragraph of MIL-E-16400. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel except in those applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on receiver front panels and electronic plug-in assemblies. Cadmium plating is not acceptable on the captive-type bolts used on the receiver front panels and electronic plug-in assemblies.

3.12.1.6 Electrical and RF connectors. Electrical and RF connectors shall conform to the Connectors, electrical and Connectors RF paragraphs of MIL-E-16400.

3.12.1.6.1 RF input and output connectors. The coaxial RF connectors for the antenna input, IF output, and external frequency standard input and output shall be rear-panel mounted. The connector for the antenna input shall be Type N in accordance with MIL-C-39012. The connector for the IF output and external frequency standard input and output shall be Type BNC in accordance with MIL-C-39012.

3.12.1.6.2 Assembly (module) connectors. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating.

3.12.1.7 Time meter. A time meter is not required.

3.12.1.8 Fuses. Fuses shall be in accordance with the Fuses and fuseholders paragraph of MIL-E-16400, except that extra fuses are not required.

3.12.1.9 Filters. Filters used in the sideband selection process shall be mechanical or crystal filter type.

3.12.1.10 Telephone jacks. Telephone jacks shall be provided on the front panel to permit the use of earphones in place of a speaker for full operation of the receiver.

3.12.1.11 Telephone plugs. Telephone plugs are not required.

3.12.1.12 RF signal strength meter. A meter shall be provided on the front panel to indicate relative RF signal strength levels.

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3.12.1.13 Assembly identification. Assemblies shall be provided with the appropriate assigned National stock number, manufacturer's part number (including revision indicator), and Federal supply code number. This information shall be inscribed on the assemblies.

3.12.1.14 Dust covers. The dust covers of all assemblies employing printed circuit boards (PCBs) shall indicate the location of all test points and all adjustments accessible through the dust cover.

3.12.1.15 Accessibility. In addition to the requirements of the Accessibility paragraph of MIL-E-16400, the circuit mounting boards and components shall be easily accessible for maintenance, repair, or replacement. Exterior doors, hinged covers, and slide drawers shall have automatic latch or quick release devices which must be actuated before they can be opened, and stops to lock them in the open position for each maintenance configuration. Sufficient clearance shall be provided in each maintenance configuration to allow required tasks to be performed without undue discomfort, danger, or effort. The chassis shall be capable of orientation ±90 degrees for maintenance purposes.

3.12.1.16 Test points. Test points shall be so provided and located as to permit the isolation of malfunctioning parts or subassemblies within any major unit or assembly without the unsoldering of any interconnection or wiring lead.

3.12.1.17 Circuit alignment. Resonant circuits shall be provided with an effective means of adjustment to permit accurate alignment. There shall be sufficient means of adjustment to compensate for the effects of normal aging of components and to facilitate normal preventive maintenance adjustments.

3.12.1.17.1 Intermediate frequency (IF) bandpass filters. Composite bandpass filters may be furnished as sealed units without trimmers.

3.12.1.17.1.1 IF output frequency. The IF output frequency shall be a nominal 455 kHz.

3.12.1.17.1.2 IF output impedance. The IF output impedance shall be 50 ohms unbalanced.

3.12.1.17.1.3 IF output level. The IF output level shall be not less than 20 millivolts (mV) for an RF input level at the automatic gain control (AGC) threshold.

3.12.1.17.1.4 IF output connector. The RF connector for the IF output test port shall be Type BNC in accordance with MIL-C-39012.

3.12.1.18 Factory adjustment controls. The design of the receiver shall not include factory or sealed adjustment controls unless specifically approved by the procuring activity.

3.12.1.19 Shock and vibration. Shock and vibration shall be as specified in 3.12.1.19.1 and 3.12.1.19.2.

3.12.1.19.1 Shock. The receiver shall be capable of withstanding the shock test for Grade A, Class I, lightweight, Type A equipment as specified in MIL-S-901.

3.12.1.19.2 Vibration. The receiver shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. During vibration testing, the receiver shall be tested in accordance with the requirements of 4.5.5.17. The identification plate requirements of MIL-STD-167-1 are not applicable.

3.12.1.20 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the Printed wiring paragraph of MIL-E-16400.

3.12.1.21 Welding. Welding shall conform to the welding requirements of MIL-E-16400.

3.12.1.22 Attachment of wires and leads. The receiver shall conform to the Attachment of wires and leads paragraph of MIL-E-16400.

3.12.1.23 Part replacement. The receiver shall conform to the Part replacement paragraph of MIL-E-16400.

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3.12.1.24 Printed circuit assembly protection. With power applied to the receiver, removal or insertion of printed circuit assemblies shall not damage the assembly or any other part of the receiver. Each assembly shall be keyed to prevent incorrect insertion.

3.12.1.25 Maintenance accessories. Extender cables or extender cards, or both, as applicable, shall be furnished as required. Electronic assemblies and PCBs that plug in to chassis or other board receptacles and on which the test points or components are not available for fault isolation, require an extender cable or card for fault isolation.

3.12.1.25.1 PCB extender cards. Extender cards shall be of sufficient height to permit access to all test points and components on the PCB for dynamic fault isolation. Extender cards shall conform to the Printed wiring paragraph of MIL-E-16400.

3.12.1.26 Thermal design. Thermal design shall be in accordance with the Thermal design paragraph of MIL-E-16400.

3.12.1.27 115 VAC power connections. 115 VAC power cable connections within the receiver shall have the conductor and color assignments specified in a through c:

a. 115 VAC return shall be white; Pin A
b. Safety ground shall be green; Pin B
c. 115 VAC high (supply) shall be black; Pin C

3.12.1.28 Power supply protection. Fault conditions ranging from open circuits to short circuits shall cause no damage to the power supply.

3.12.2 Human engineering. The contractor shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein except that the receiver shall be designed for use by the 5th percentile female to the 95th percentile male.

3.12.2.1 Human error design protection. Front panel control functions shall be of such design that neither incorrect adjustment nor random sequencing of functions will cause damage to the receiver.

3.13 Detailed performance. Detailed performance shall be as specified in 3.13.1 through 3.13.18.

3.13.1 Sensitivity and gain. Sensitivity (see 6.4.7) and gain requirements shall be as specified in 3.13.1.1 through 3.13.1.3.

3.13.1.1 Sensitivity. Receiver sensitivity shall conform to the requirements specified in TABLE II and TABLE III for any frequency within the complete frequency range of the receiver. Specified sensitivities shall be achieved in all modes with the manual RF gain control in the extreme clockwise (maximum gain) position and shall be measured from RF input to audio output.

TABLE II. Sensitivity (200 kHz to 30 MHz).

Mode	CW	AM	USB, LSB, ISB	FM
IF bandwidth (kHz)	0.3	6.0	3.0	16
Sensitivity (μV) 1/	0.4	2.5	0.6	2.5
Percent mod 2/, 400 Hz	N/A	50	N/A	N/A
Deviation (kHz), 400 Hz mod	N/A	N/A	N/A	4.8
(S+N)/N 3/ (dB) 4/	16	10	10	17

1/ Microvolts
2/ Modulation
3/ Signal plus noise-to-noise
4/ Decibels

TABLE III. Sensitivity (14 kHz to 200 kHz).

Mode	CW	USB
IF bandwidth (kHz)	0.3	3.0
Sensitivity (μV)	2.0	3.0
(S+N)/N (dB)	16	10

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3.13.1.2 Gain. The receiver gain shall be sufficient to produce standard audio output (see 6.4.3) for the input voltages of TABLE II and TABLE III, for any frequency within the complete frequency range (14 kHz to 30 MHz) of the receiver.

3.13.1.3 Gain variation. The variation in overall gain for any mode of reception shall not exceed 10 dB for the complete working frequency range of each MHz frequency band. Gain variations over the complete receiver tuning range (14 kHz to 30 MHz) shall not exceed 6 dB total.

3.13.2 Selectivity. Selectivity (see 6.4.8) requirements shall be as specified in 3.13.2.1 through 3.13.2.4.

3.13.2.1 Selectivity (antenna input to detector input) (see 6.4.8.1). The selectivity of the receiver, excluding the audio frequency (AF) response characteristics, shall depend primarily upon the characteristics of the filter defining the final IF passband. When measured in accordance with 4.5.5.2, selectivity shall be as specified in 3.13.2.1.1, 3.13.2.1.2, and 3.13.2.1.3.

3.13.2.1.1 USB, LSB, and ISB operation. USB, LSB, and ISB operation shall be as specified in a through h:

a. USB passband (-3 dB): Carrier +450 Hz to carrier +3000 Hz
b. LSB passband (-3 dB): Carrier -450 Hz to carrier -3000 Hz
c. ISB passband (-3 dB): Carrier +450 Hz to carrier +3000 Hz, and carrier -450 Hz to carrier -3000 Hz
d. Ripple (peak to valley): 3 dB maximum
e. Attenuation at carrier frequency: 10 dB minimum
f. Opposite sideband attenuation 1 kHz into opposite sideband: 60 dB minimum
g. Attenuation at carrier +5300 Hz (USB): 60 dB minimum
h. Attenuation at carrier -5300 Hz (LSB): 60 dB minimum

3.13.2.1.2 AM operation. AM operation shall be as specified in a through c:

a. Passband (-3 dB): Carrier -3000 Hz to carrier +3000 Hz
b. Ripple (peak to valley): 3 dB maximum
c. Attenuation at carrier +9.5 kHz, carrier -9.5 kHz: 60 dB minimum

3.13.2.1.3 CW operation. The -3 dB CW passband shall be 300 Hz. The -60 dB passband shall not exceed 1.0 kHz.

3.13.2.2 Overall passband (antenna input to audio output) (see 6.4.8.2). When measured in accordance with 4.5.5.2.1, the overall carrier passband characteristics shall be as specified in 3.13.2.2.1.

3.13.2.2.1 USB, LSB, and ISB characteristics. USB, LSB, and ISB receiver passband characteristics shall be as specified in a through d:

a. USB passband (-3 dB): Carrier +450 Hz to carrier +3000 Hz
b. LSB passband (-3 dB): Carrier -450 Hz to carrier -3000 Hz
c. ISB passband (-3 dB): Carrier +450 Hz to carrier +3000 Hz, and carrier -450 Hz to carrier -3000 Hz
d. Ripple (peak to valley):
    1. Carrier +450 Hz to carrier +3000 Hz: 3 dB maximum USB
    2. Carrier -450 Hz to carrier -3000 Hz: 3 dB maximum LSB

3.13.2.3 Reciprocal mixing. With the receiver in the SSB mode, 3 kHz bandwidth, and an unmodulated resonant wanted signal of -107 decibels referred to one milliwatt (dBm), the output (S+N)/N ratio shall be greater than or equal to 10 dB with an unwanted signal at a Δf greater than or equal to 100 kHz and a level of -17 dBm.

3.13.2.4 Blocking. With the receiver in SSB mode, 3 kHz bandwidth, an unmodulated resonant signal of -77 dBm, and the receiver gain adjusted so that the wanted signal is just below the AGC threshold, the output (S+N)/N ratio shall be greater than or equal to 10 dB and the receiver output level shall not be reduced by more than 3 dB when an unwanted signal at a Δf of greater than or equal to 100 kHz and a level of +12 dBm is introduced.

3.13.3 Spurious responses (see 6.4.9). The receiver shall conform to the requirements specified in 3.13.3.1 through 3.13.3.6.

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3.13.3.1 Internally generated spurious outputs. With no RF input signal, the antenna input terminated with a shielded 50-ohm resistor, and the receiver adjusted for the reception of SSB signal, internally generated receiver outputs shall conform to the requirements specified in 3.13.3.1.1 and 3.13.3.1.2.

3.13.3.1.1 Receiver operated on an internal frequency standard. No more than seven spurious responses shall occur with output levels equivalent to that resulting from a 0.4 µV to 4.0 µV (open circuit) reference signal applied to the antenna input. All other spurious responses generated within the equipment shall have output levels less than that resulting from a 0.4 µV (open circuit) reference signal applied to the antenna input.

3.13.3.1.2 Receiver operated on an externally supplied 1 MHz reference. When, in lieu of operating with the internal frequency standard, the receiver is supplied an external 1 MHz reference at an available power of +21 dBm (5 V open circuit in series with 50 ohms), having all nonharmonically related components at least 90 dB, and all harmonically related components at least 60 dB below the reference, internally generated spurious outputs shall conform to the requirements of 3.13.3.1.1.

3.13.3.2 Image rejection. Image rejection (see 6.4.9.1), measured at the audio output, shall be not less than 80 dB.

3.13.3.3 IF rejection (see 6.4.9.2). Rejection of all IFs, measured at the audio output, shall be not less than 80 dB.

3.13.3.4 Other single-signal spurious responses. At frequencies more than ±20 kHz from the tuned frequency, the spurious signal rejection shall be at least 80 dB.

3.13.3.5 Front-end intermodulation (out-of-band). CW signals at two frequencies, F1 and F2, shall be applied simultaneously to the antenna input from a 50-ohm linear combiner. Signal pairs shall be (F1, F2) = (f_o + Δf, f_o + 2 Δf) and (f_o - Δf, f_o - 2 Δf), where f_o is the resonant frequency of the sideband and Δf is greater than 0. The signal pairs shall be increased in amplitude, maintaining equality of the two signals until the receiver output reaches 10 dB (S+N)/N or until each input signal amplitude reaches the specified levels for that pair. The specified levels for the first pair shall be -20 dBm for Δf greater than or equal to 100 kHz and -5 dBm for Δf greater than or equal to $0.5f_o$. The levels for the second pair shall be -20 dBm for Δf greater than or equal to 100 kHz and -5 dBm for Δf greater than or equal to $0.3f_o$. The level for the third pair shall be +20 dBm for Δf greater than f_o.

3.13.3.6 Cross mod. With the receiver in the SSB mode, 3 kHz bandwidth, and an unmodulated signal at the receiver resonant frequency at a level of -77 dBm, the desired output shall be modulated less than 10 percent (cross mod sidebands less than 26 dB) by an unwanted 30 percent amplitude modulated signal at a frequency differing from the receiver frequency by greater than or equal to 100 kHz, and a level of +12 dBm.

3.13.4 Audio outputs. The audio outputs from the USB and LSB amplifiers shall conform to the requirements of 3.13.4.1 through 3.13.4.4.2. Each amplifier shall feed line output receptacles at the rear of the receiver and a head telephone jack on the front panel. The line outputs shall terminate in appropriately identified receptacles.

3.13.4.1 Audio output voltage regulation. Each audio amplifier shall be so designed that the output voltage shall remain constant within 30 percent (in V) of the value obtained with a 600-ohm load for a change in the value of this load from 100 ohms to 600 ohms in 100-ohm steps. This condition shall be obtained with any input level from zero input to that which produces full output. The power output transistors shall not be subjected to damage or deterioration at any output load impedance from zero ohms to an open circuit.

3.13.4.2 Maximum undistorted output. With an RF input of not less than 1000 µV, the maximum undistorted output (see 6.4.11) shall be not less than 100 milliwatt (mW), dissipated in a 600-ohm resistive load at the line output receptacles of the audio amplifier; or with the line level control set at the maximum rated output setting, the undistorted output appearing across a 600-ohm resistive load connected to the telephone jack shall be not less than 30 mW. In the AM mode, with 30 percent mod, distortion shall be not greater than 5 percent.

3.13.4.3 Audio level controls. Separate (noncoaxial) level controls shall be provided on the receiver front panel for audio line and local (telephone) outputs. Telephone level controls shall be equipped with knobs chosen in accordance with the part selection procedures specified in MIL-E-16400. Line level controls shall be provided on the front panel for audio line controls and shall be a screwdriver-adjustable type. Controls shall be standard audio (clockwise log) taper, and shall provide smooth variation in output.

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3.13.4.3.1 Audio level metering. The receiver shall be provided with a front panel audio metering system which will indicate relative audio line output levels.

3.13.4.4 Output hum and noise. Output hum and noise shall be as specified in 3.13.4.4.1 and 3.13.4.4.2.

3.13.4.4.1 Output signal-to-hum ratio. The output signal-to-hum ratio for AM, USB, or LSB shall be not less than 45 dB for the standard audio output (see 6.4.3) of 1 mW. This requirement shall apply separately to each output component which is related to the power line frequency, including hum sidebands on the desired audio output. Measurement shall be in accordance with 4.5.5.4.5.

3.13.4.4.2 Output signal-to-hum-plus-noise ratio. The output signal-to-hum-plus-noise ratio for AM, USB, and LSB modes, measured in accordance with 4.5.5.4.5, shall be not less than 35 dB.

3.13.5 Manual gain control. The manual gain control shall have the range to provide a variation in the overall gain of the RF and IF amplifiers of the receiver of not less than 100 dB. The gain control shall be accessible from the front panel and shall be disabled in the AGC mode.

3.13.6 AGC characteristics. The AGC characteristics shall be as specified in a and b:

a. Audio output shall remain constant within a total tolerance of 3 dB for an increase of 100 dB above AGC threshold.
b. The AGC threshold (see 6.4.14) shall be adjustable between 1.0 µV and 5.0 µV.

3.13.6.1 AGC conditions. The requirements of 3.13.6 shall apply for the conditions specified in a through d, singly or in any combination:

a. Any frequency selected within the nominal frequency range of the receiver
b. Any audio gain adjustment producing audio levels at or below maximum undistorted output
c. Any constant level of carrier mod between 30 percent and 90 percent
d. Any constant mod frequency between the limits of 450 Hz and 3000 Hz

3.13.6.2 AGC time constants. The AGC time constants (see 6.4.13) shall be as specified in TABLE IV. The time constants shall hold for signal transitions of 60 dB or less.

TABLE IV. AGC time constants.

AGC	Attack time	Decay time
FAST	<20 ms[1/]	<35 ms
SLOW	<20 ms	5 seconds maximum

1/ Milliseconds

3.13.6.3 Resonant overload. Audio output power (remote outputs) at resonant overload (see 6.4.10) shall be a minimum of 100 mW.

3.13.6.4 Time constant of recovery from surge inputs. The design shall be such that with the receiver operating under standard conditions as for the reception of CW signals and with AGC off, a sharp pulse of CW input signal, at resonance, having a minimum amplitude of 5 V and duration of 50 ms (to simulate static surge), will not cause a reduction in sensitivity that will persist in excess of 40 ms after cessation of the input surge.

3.13.6.4.1 Measurement of time constant. The time constant shall be measured from the time of removal of the input signal pulse until the output noise level has reached 63 percent of the steady-state value. Start of time shall begin at the moment of removal of the input signal pulse.

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3.13.7 Beat frequency oscillator (BFO) control. The BFO control shall operate (when the mode is set at CW) so as to permit a variation of the frequency of the BFO of not less than ±3000 Hz.

3.13.8 In-band intermodulation. Intermodulation distortion products generated when the receiver processes two equal-amplitude unmodulated RF signals producing audio outputs at 935 Hz and 1045 Hz shall each be a minimum of 40 dB below the reference output tones. This requirement shall be verified in USB and LSB operation with the AGC time constant selector switch in both FAST and SLOW positions, over an input amplitude range of 100 µV to 100,000 µV, open circuit, per tone, and for audio output levels of 0 dBm, or less, per tone.

3.13.9 RF input circuit. The receiver shall not be damaged by an input signal of up to 70 volts root-mean-square (Vrms) and shall recover to full performance within 40 ms after removal of this signal. The degree of protection shall exist with the receiver on or off. The threshold of operation of a protection circuit shall be greater than 2.0 peak envelope voltage.

3.13.10 Accuracy of main tuning indicator calibration. The accuracy of frequency indications shall be within the limits specified in 3.13.10.1. The specified accuracies shall apply to all tuning indicators, acting in combination, to indicate the tuning of the receiver.

3.13.10.1 Tuning steps. Tuning in steps of 10 Hz or less shall be provided with an accuracy equivalent to that of the internal frequency standard. The synthesizer lock or settling time shall not exceed 100 ms.

3.13.11 Frequency and gain stability. The receiver shall include provisions for operation from either an internal or external frequency standard of 1 MHz. All frequency stability requirements apply to operation using the internal standard.

3.13.11.1 Frequency stability and gain variations. The requirements of 3.13.11.1.1 through 3.13.11.1.6.2 shall apply under conditions of CW or sideband reception (see 6.4.2.1), as specified. The frequency stability figures shall apply to the variation in a 1000 Hz AF beat note with a constant input frequency. The frequency changes shall be made with reference to the frequency variations of all oscillators on the audio beat note frequency. All measurements shall refer to total variations, either instantaneous, transient, or gradual changes. The requirements specified herein shall apply at any frequency within the frequency range of the receiver.

3.13.11.1.1 Sixty-minute warm-up period. Measurements shall be made in the sideband mode at constant ambient temperature, constant relative humidity and constant line input voltage. Two minutes after turn-on time, the receiver shall be within 0.1 percent of the frequency it will have reached at the end of 60 minutes. Fifteen minutes after turn-on time, the receiver shall be within 0.002 percent of the frequency reached 60 minutes after turn-on. Sixty minutes after turn-on, the receiver shall be within one part in 10^7 of the desired frequency.

3.13.11.1.2 Variation in primary power voltage and frequency. Variation of primary power supply line voltage over a range of 115 V, ±10 percent and of power line frequency over a range of 58 Hz to 62 Hz, in any combination, shall not cause a change in the frequency stability of the receiver of more than two parts in 10^8 (total), or a change in overall gain (AGC defeated) in excess of 2 dB.

3.13.11.1.3 Variation in relative humidity. Exposure of the receiver to humidity conditions specified in 3.9 shall not cause loss of synthesizer frequency lock or excessive changes in frequency or gain.

3.13.11.1.4 Variation in ambient temperature. The receiver, when in the operating condition and when subjected to the operating temperature extremes for Range 4 equipment specified in MIL-E-16400, shall not exhibit frequency or gain variation in excess of the limits shown in TABLE V. The specified limits for variations of frequency and gain shall be the total deviations of these parameters, which occur during the time period required by the high and low operating temperature cycling test procedures specified in MIL-E-16400. The AGC shall be off during gain variation measurements.

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TABLE V. Frequency and gain variation.

Performance requirement	Variation
Frequency variation versus change in ambient temperature from 0°C[1/] to +50°C (32°F[2/] to 122°F)	1 part in 10^7
Gain variation versus change in ambient temperature from 0°C to +50°C (32°F to 122°F)	4 dB (total)

1/ Celsius
2/ Fahrenheit

3.13.11.1.5 Control variations. Frequency variations resulting from a variation of any nonfrequency-determining control throughout the full range of operation shall not exceed five parts in 10^8. This requirement shall apply to those controls which operate in nonfrequency-determining circuits, and to controls which affect the RF or AF gain or level. The specified limit for frequency variation shall be the absolute maximum deviation of this parameter which occurs during the manipulation of any control throughout its full range of operation.

3.13.11.1.6 Short and long term stability. The receiver shall be capable of operation for 24 hours without operator attention. Without operator attention shall be interpreted as meaning that reliable reception of a desired signal shall be assured throughout a 24-hour period without the necessity for readjustment of any tuning control. This requirement shall be applicable throughout the complete frequency range of the receiver.

3.13.11.1.6.1 Sudden changes during 24-hour period. During the 24-hour period of operation specified in 3.13.11.1.6, the receiver shall not exhibit sudden sporadic shifts in the frequency generated by any of the oscillators.

3.13.11.1.6.2 Frequency drift rate. After an initial stabilization period, not to exceed 48 hours following turn-on, the overall frequency drift rate shall not exceed 1 part in 10^8 (total) per day. This rate shall be achieved throughout the receiver operating frequency range in any mode of operation which does not use the BFO or frequency vernier. The long-term frequency drift rate, averaged over a period of 60 days, shall not exceed 3 parts in 10^9 (total) per day. During measurement of frequency drift rate, changes in power supply line voltage and frequency shall be limited to ±1 percent; temperature variations shall be limited to 10°C (50°F) (total). Under the same conditions, receiver gain (AGC off) shall not change by more than 1 dB (total) during a 24-hour period after initial stabilization.

3.13.12 Regeneration. There shall be no evidence of unintended oscillations or instability under any condition of receiver operation, including primary line voltage 10 percent higher than the nominal primary line voltage.

3.13.13 Oscillator radiation. Oscillator radiation shall conform to the Class A4 requirements of MIL-STD-461. Antenna terminal narrowband emissions in the region 10 kHz to 1 gigahertz (GHz) shall not exceed -90 dBm. Broadband emissions in the same frequency range shall comply with the requirements of Method CE06, MIL-STD-461.

3.13.14 Frequency standard adjustment range. The internal 1 MHz frequency standard shall be capable of ±20 Hz adjustment and a setting resolution of 1 part in 10^9. The adjustment shall be accessible from the top surface of the frequency standard assembly without removing the assembly cover. Piston type adjustments shall be protected from damage at the ends of their adjustment ranges.

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3.13.14.1 Operation using an external frequency standard. The receiver shall be capable of operation from an external frequency standard of 1 MHz with any input level between 400 mV and 10.0 V, open circuit, in series with 50 ohms. Input impedance at the external 1 MHz input terminals shall be 50 ohms, nominal, within a voltage standing wave ratio of 2.0:1 or less.

3.13.14.2 Reference output signal. The receiver shall provide a 1 MHz reference output signal of between +7 dBm and +13 dBm, delivered to a 50-ohm load, for testing.

3.13.14.3 Frequency standard switch functions. The internal frequency standard assembly shall have a function switch accessible without removal of the assembly dust cover. This switch shall have two positions with the functions specified in a and b:

a. EXT: Permits operation from an external frequency source. The internal frequency standard oscillator and oven shall be kept on for immediate availability with full accuracy.

b. INT: Permits operation from the internal standard.

3.13.15 Frequency generation. The frequency generation process shall utilize only a single reference frequency, with all internally generated frequencies used in the synthesis of the operating frequency derived by arithmetic operations on, and phase-locked to the reference.

3.13.16 Electromagnetic compatibility (EMC). The receiver shall conform to the requirements of Class A4, MIL-STD-461.

3.13.16.1 Electromagnetic pulse (EMP). The LF, MF, and HF receiver shall be hardened or shall be intrinsically hard against the EMP resulting from an exo-atmospheric nuclear burst to survive the levels specified in a and b:

a. Receivers operated external to an intentionally hardened area shall not exhibit permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances and recovery times specified in the individual equipment or subsystem specification after being subjected to a test signal having the waveform and amplitude shown in FIGURE 1.

b. Receiver and subsystem interface pins and terminals of leads, including grounds and neutrals shall not exhibit permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances and recovery times specified in the individual equipment or subsystem specification after being subjected to a test signal having either the waveform and common mode currrent level shown in FIGURE 2 or the wave form and common mode voltage level shown in FIGURE 3, whichever occurs first, at the specific test frequency.

3.13.17 Remote operation. The receiver shall be capable of remote control operation. The functions specified in a through e shall be controlled by the remote unit:

a. IF bandwidth
b. Mode selection (AM, FM, CW, USB, LSB, ISB)
c. AGC (FAST-SLOW) or manual gain control
d. Tune frequency
e. BFO control

3.13.17.1 Remote electrical interface. The electrical interface for remote operation shall conform to MIL-STD-188.

3.13.18 Carrier operated relay (COR). Means shall be provided to permit the operator to select a relative signal strength for operation of a relay. The relay contacts shall be brought out at the rear panel.

3.14 Safety criteria. Receiver design shall conform to the Safety criteria paragraph of MIL-E-16400, except as specified herein.

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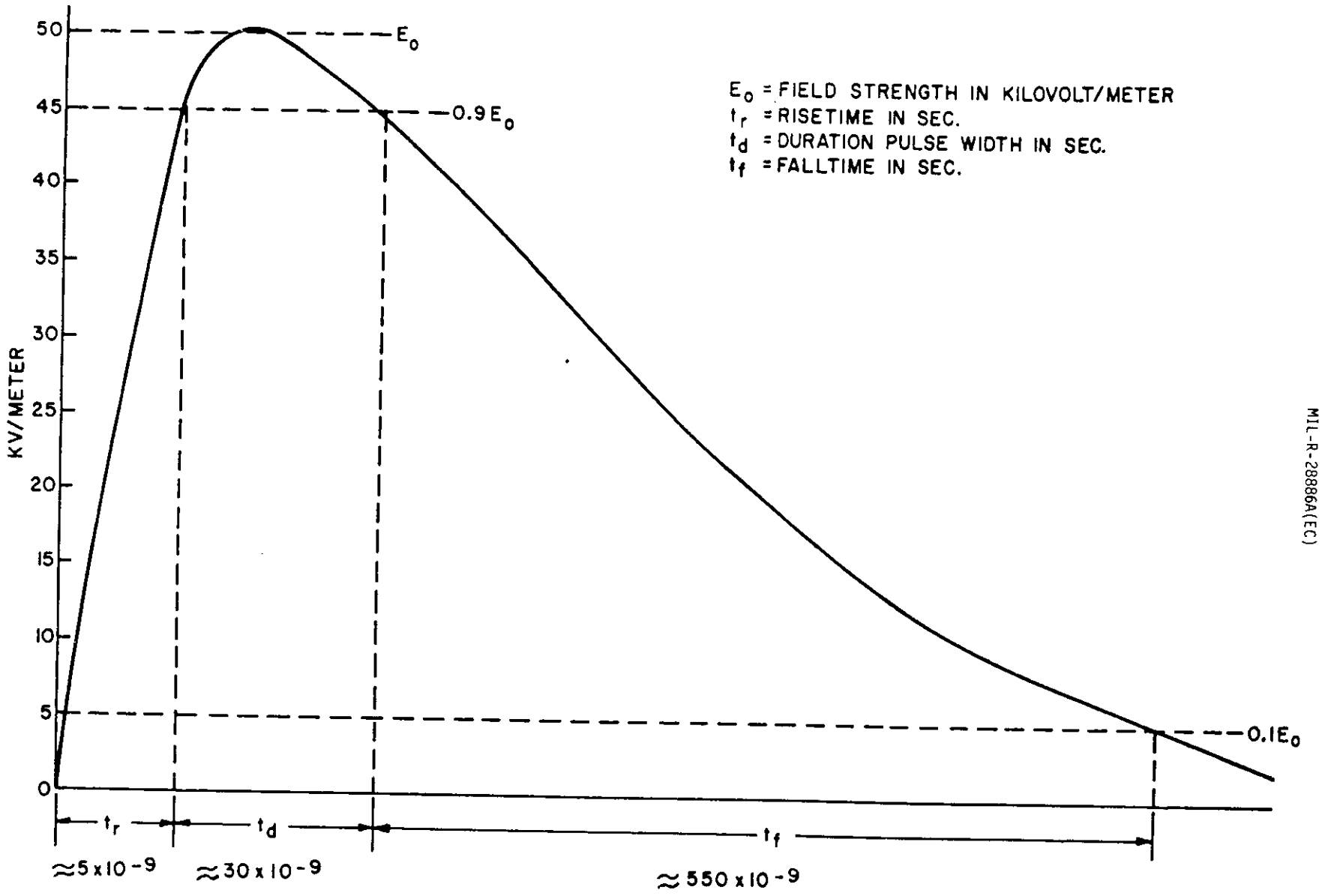
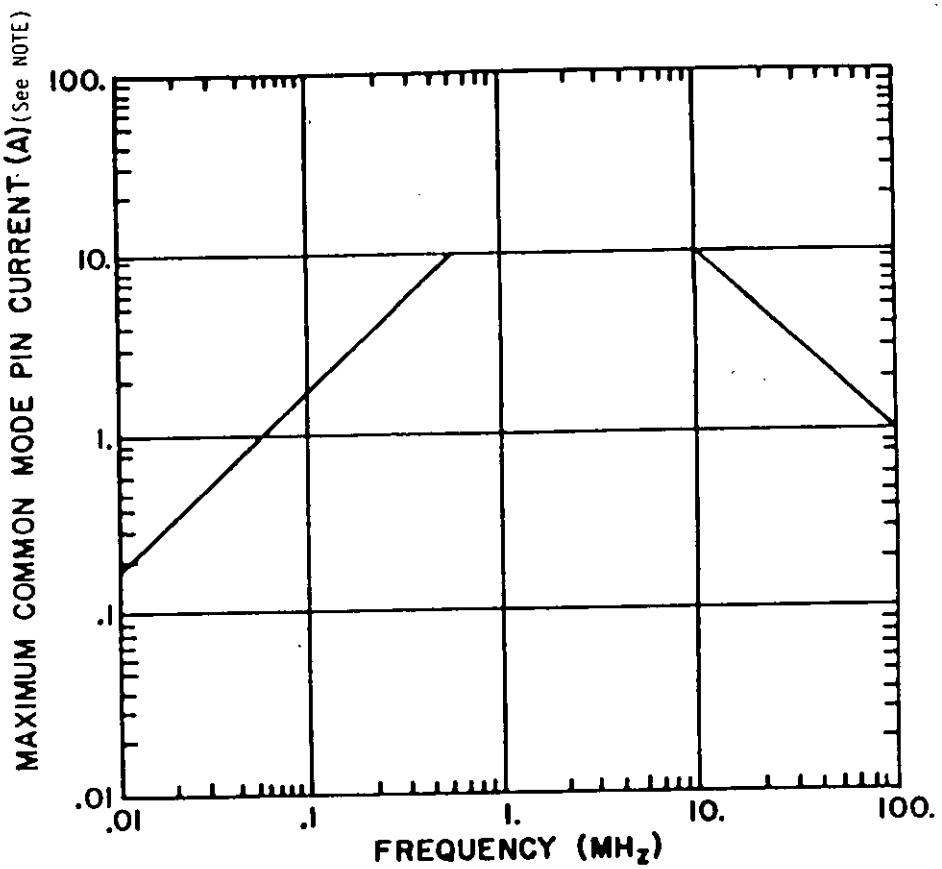
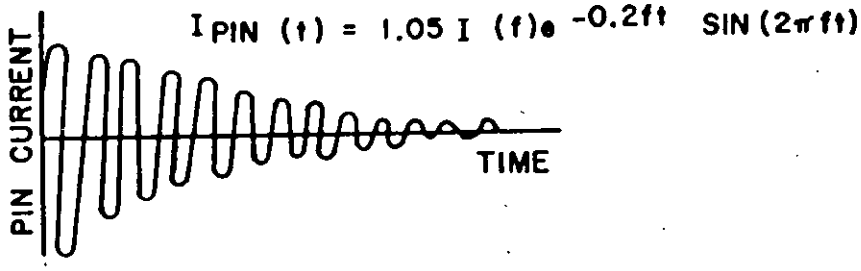


FIGURE 1. EMP test signal.

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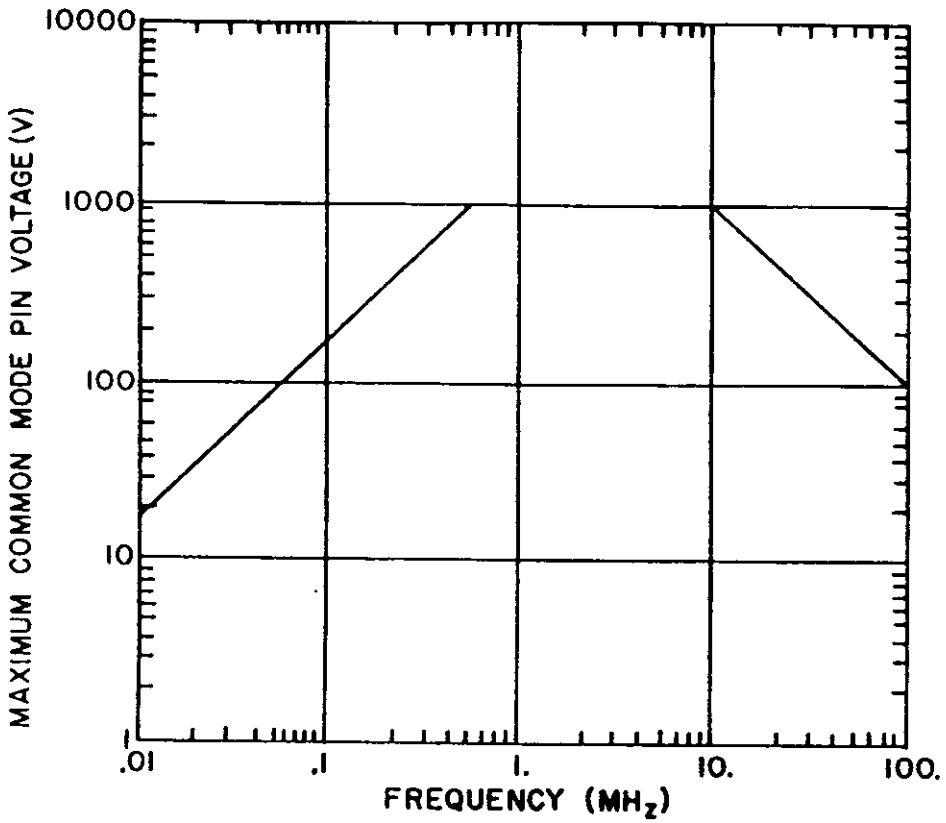


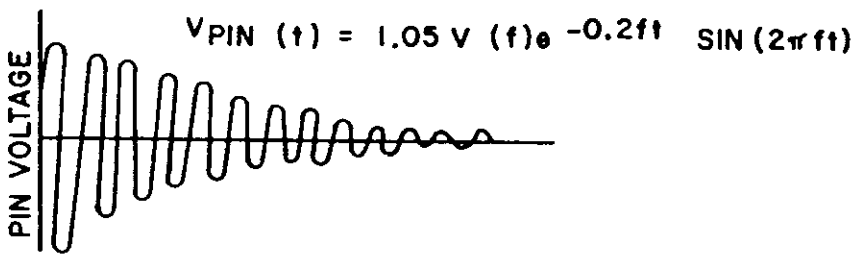
NOTE: Amperes

Interface pin current

FIGURE 2. EMP common mode current level.

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Interface pin voltage

FIGURE 3. EMP common mode voltage level.

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3.15 Identification and marking. Identification and marking of all units, assemblies, electrical and moving mechanical parts, and mechanical parts which require removal for normal maintenance shall be in accordance with the Identification and marking paragraph of MIL-E-16400 and MIL-E-21981. Hertz or Hz shall be used for all indications of frequency on hardware and associated documentation.

3.15.1 Weight marking. Receivers weighing in excess of 15.87 kg (35 lbs) shall have the weight clearly marked on the external surface of the receiver and readily visible during installation and removal. The marking shall be as permanent as the normal life expectancy of the unit to which it is applied.

3.16 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined in MIL-STD-781) shall be 5000 hours. The lower test MTBF (θ_1 as defined in MIL-STD-781) shall be 2500 hours.

3.17 Maintainability. Maintainability shall be as specified in 3.17.1 and 3.17.2.

3.17.1 Quantitative maintainability requirement. The receiver shall have a mean-time-to-repair (MTTR) not exceeding 0.5 hour and a maximum-corrective-maintenance-time (M_{maxct}) at the 95th percentile not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The MTTR includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The receiver is in a corrective maintenance mode whenever a failure, including a malfunction that allows degraded operation, occurs.

3.17.2 Built-in test (BIT) feature. The receiver shall contain a BIT feature which shall be capable of fault detection and isolation at the level of the lowest replacement module and shall report operational status with both local and remote display.

3.18 Workmanship. Workmanship shall be as specified in 3.18.1 and 3.18.2.

3.18.1 General workmanship. The workmanship and soldering requirements shall be in accordance with the Workmanship and Soldering paragraphs of MIL-E-16400. Printed circuit assemblies shall be in accordance with MIL-I-46058 and MIL-P-28809.

3.18.2 Workmanship screen. Receivers shall withstand a temperature cycling test as specified in 4.9.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality and safety assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality and safety programs are being properly applied; b) Government product inspection to measure quality and safety of the product to be offered for acceptance; and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Safety verification (see 4.4.5)
d. Inspection of preparation for delivery (see 4.10)

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4.3 First article inspection. Unless otherwise specified (see 6.2.1), three units shall be required for first article inspection. First article inspection shall consist of all examinations and tests necessary to determine compliance with the requirements of this specification. First article inspection shall include the examinations and tests specified in TABLE VI.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.5.

4.4.1 Production inspection (Group A). Production inspection shall be performed on every receiver offered for delivery. Production inspection shall encompass such examinations and tests which will prove the workmanship and reveal any omissions or errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls or adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests shown in Group A of TABLE VI, subject to the conditions of 4.4.1.1.

4.4.1.1 Group A tests. The Group A tests shall be performed in accordance with the test methods of 4.5.4 except as modified by a through f:

a. Image rejection: HF mixer image only
b. IF rejection: Of the first IF only
c. Maximum undistorted output: 2.1 MHz
d. Output signal-to-hum ratio: 2.1 MHz
e. AGC characteristics: 2.1 MHz
f. Frequency stability: 5 MHz

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in MIL-STD-105 using special inspection level S-3 for normal, tightened, and reduced inspection. The acceptable quality level shall be 6.5 percent defective for each of the Group B examinations and tests. Production control inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; and tests to determine temperature rise produced in operation and ability of the receiver to withstand this heat. These tests shall be performed on the complete receiver as offered for delivery. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using a more stringent inspection. Rejected lots shall be kept separate from the new lots and shall not be allowed to lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis, as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the receiver as a whole; life tests, simulated service tests; tests of the effects of changes in environment (such as extremes of temperature and humidity or effect of salt air), and tests of the effects of shock and vibration. Environmental inspection shall include the examinations and tests specified in Group C of TABLE VI.

4.4.3.1 Sampling for environmental inspection. One receiver from the first month's production lot, or from the first 50 units produced, shall be subjected to environmental inspection. One receiver from each successive production lot of 100 shall be subjected to environmental inspection. One receiver from each 50 receivers produced shall be subjected to electromagnetic interference (EMI) tests. Environmental inspection shall be performed on receivers that have passed the production inspection of 4.4.1.

4.4.3.2 Sample failure. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

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TABLE VI. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Weight	3.8	4.5.1	X		X	
Dimensions	3.8	4.5.1	X		X	
Parts and materials	3.1.2	4.5.1	X		X	
Marking	3.15	4.5.1	X	X		
Welding	3.12.1.21	4.5.1	X	X		
Mechanical design and construction	3.12	4.5.1	X	X		
Bonding and grounding	3.12.1.3	4.5.1	X		X	
Corrosion protection	3.12.1.4	4.5.1	X		X	
Painting	3.12.1.4	4.5.1	X		X	
Assembly identification	3.12.1.13	4.5.5.22	X		X	
Soldering	3.18.1	4.5.1	X		X	
Frequency range	3.3.1	4.5.5.10	X	X		
Overcurrent protection	3.4.2	4.5.1	X		X	
Leakage current	3.4.3	4.5.5.26	X		X	
Sensitivity	3.13.1.1	4.5.5	X	X		
Gain	3.13.1.2	4.5.5.1	X	X		
Gain variation	3.13.1.3	4.5.5.1.1	X	X		
Selectivity	3.13.2	4.5.5.2	X	X		
Overall passband	3.13.2.2	4.5.5.2.1	X	X		
Reciprocal mixing	3.13.2.3	4.5.5.2.2	X		X	
Blocking	3.13.2.4	4.5.5.2.3	X		X	
Spurious responses	3.13.3	4.5.5.3	X		X	
Internal spurious responses	3.13.3.1	4.5.5.3.1	X	X		
Image rejection	3.13.3.2	4.5.5.3	X	X		
IF rejection	3.13.3.3	4.5.5.3	X	X		
Single signal responses	3.13.3.4	4.5.5.3	X	X		
Front end intermodulation	3.13.3.5	4.5.5.3	X		X	
Cross mod	3.13.3.6	4.5.5.2.4	X		X	
Audio output voltage regulation	3.13.4.1	4.5.5.4.1	X		X	
Maximum undistorted output	3.13.4.2	4.5.5.4.2	X	X		
Audio level controls	3.13.4.3	4.5.5.4.3	X		X	
Audio level metering	3.13.4.3.1	4.5.5.4.4	X		X	
Signal-to-hum ratio	3.13.4.4.1	4.5.5.4.5	X	X		
Signal-to-hum and noise	3.13.4.4.2	4.5.5.4.5	X		X	
Manual gain control	3.13.5	4.5.5.5	X		X	
AGC characteristics	3.13.6	4.5.5.6	X	X		
AGC time constants	3.13.6.2	4.5.5.6.1	X		X	
COR	3.13.18	4.5.4	X	X		
Resonant overload	3.13.6.3	4.5.5.6.2	X		X	
Surge recovery	3.13.6.4	4.5.5.6.3	X		X	
BFO control	3.13.7	4.5.5.7	X		X	
In-band intermodulation	3.13.8	4.5.5.8	X		X	
RF input circuit	3.13.9	4.5.5.9	X		X	
Accuracy of tuning	3.13.10	4.5.5.10	X		X	
Tuning time	3.13.10.1	4.5.5.10.1	X		X	
Frequency stability	3.13.11	4.5.5.11	X			X
Gain stability	3.13.11	4.5.5.11	X			X
Sixty-minute warm-up period	3.13.11.1.1	4.5.5.11.1	X		X	
Line voltage variation	3.13.11.1.2	4.5.5.11.2	X			
Line frequency variation	3.13.11.1.2	4.5.5.11.2	X			
Variation in relative humidity	3.13.11.1.3	4.5.5.11.3	X			
Variation in ambient temperature	3.13.11.1.4	4.5.5.11.4	X			
Control variations	3.13.11.1.5	4.5.5.11.5	X		X	
Frequency drift rate	3.13.11.1.6.2	4.5.5.11.3	X			X
Regeneration	3.13.12	4.5.5.12	X		X	

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TABLE VI. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Oscillator radiation	3.13.13	4.5.5.13	X		X	
External frequency standard	3.13.14.1	4.5.5.14.1	X		X	
Frequency standard switch functions	3.13.14.3	4.5.4	X		X	
Steady-state voltage switch	3.4	4.5.5.18	X		X	
Power and power factor	3.4	4.5.5.20	X		X	
Steady-state frequency	3.4	4.5.5.18	X		X	
Transient voltage	3.4.1	4.5.5.18	X		X	
Transient frequency	3.4.1	4.5.5.18	X		X	
Power interruption	3.4.1	4.5.5.19	X		X	
Temperature sequence	3.9	4.5.5.16	X			X
Low temperature	3.9	4.5.5.16	X			X
High temperature	3.9	4.5.5.16	X			X
Humidity	3.9	4.5.5.16	X			X
Salt fog (spray)	3.9	4.5.1	X			X
Reliability	3.16	4.7				
Performance conditions	3.11	4.6	X			
Shock	3.12.1.19.1	4.5.5.17	X			X
Vibration	3.12.1.19.2	4.5.5.17	X			X
Thermal design	3.12.1.26	4.5.1	X			X
EMC	3.13.16	4.5.5.23	X			X
EMP	3.13.16.1	4.5.5.23.1, 4.5.6	X			X
Remote operation	3.13.17	4.5.5.24	X		X	
Maintainability	3.17	4.8	X			
General workmanship	3.18.1	4.9.1	X	X		
Workmanship screen	3.18.2	4.9.2	X	X		
Safety	3.14	4.4.5	X	X		
Chassis orientation	3.12.1.15	4.5.5.21	X	X		

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to and have passed production control or environmental inspections, or both, may be accepted on the contract provided that they are resubjected to, and pass, production inspection after repair of all defects (see 6.6).

4.4.5 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.14.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.6.

4.5.1 Examinations and tests. Examinations and tests specified in TABLE VI shall be conducted in accordance with the applicable test methods of MIL-E-16400 and MIL-STD-415, except as otherwise specified in 4.5.2 through 4.5.6. Where 4.5.1 is specified in TABLE VI, the test methods specified in MIL-E-16400 and MIL-STD-415 shall be applied.

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4.5.2 Cables. Interconnecting cables to be supplied with the end item shall be installed on the end item prior to any testing, and shall be used for the duration of all testing.

4.5.3 Contractor validation. The contractor shall inspect the receiver to determine technical compliance and adequacy for the specified characteristic or requirement. This validation shall include qualitative examination and testing whenever a function is specified.

4.5.4 Inspection by testing. When this paragraph is specified in TABLE VI, the contractor shall perform qualitative testing.

4.5.5 Sensitivity. Measurements shall be made at the absolute band edges in each of the primary frequency bands for the USB mode, and at the frequency of poorest USB sensitivity for USB, AM, and CW modes.

4.5.5.1 Gain. Measurement shall be made at the frequency of lowest gain as determined by the procedure specified in 4.5.5.1.1.1.

4.5.5.1.1 Gain variation. Measurements shall be made using the procedure specified in 4.5.5.1.1.1 to determine the gain variation of each MHz frequency band. Measurements shall be made with the receiver in USB mode, with AGC off.

4.5.5.1.1.1 Gain variation measurement procedure. Apply a signal of one mV, open circuit, in series with 50 ohms, at a frequency of 2.1010 MHz through the receiver antenna input terminal. Tune the receiver to 2.1000 MHz in USB mode. Adjust the RF gain control for maximum and turn AGC to OFF. Adjust the audio line level control to produce 1 mW of audio output. Leaving the gain and level controls and the input signal amplitude fixed, tune the receiver and signal source together to produce a constant output frequency of 1 kHz. Within each MHz frequency band, each setting of the 100 kHz shall be tested. Repeat the procedure, determining the minimum output in each band.

4.5.5.2 Selectivity (antenna input to detector input). Measurements shall be made in USB, LSB, and AM modes with the receiver tuned to 17.5500 MHz. AGC shall be inoperative and RF gain shall be maximum. Measurements shall be made of all parameters specified in 3.13.2.1.1 and 3.13.2.1.2, using an unmodulated test signal. Ripple in the passband shall be measured between the -3 dB points using a constant amplitude RF input signal which produces an output (S+N)/N ratio of 20 dB at the peak of the passband ripple. The amplitude and frequency (relative to the carrier) of each inflection point shall be determined. Measurements of carrier and opposite sideband rejection and 60 dB attenuation points shall be made by increasing the RF input signal amplitude to obtain the reference IF output produced at the peak of passband ripple.

4.5.5.2.1 Overall passband (antenna input to audio output). Measurements shall be made in USB and LSB modes, with the receiver tuned to 17.5500 MHz. AGC shall be inoperative. Bandwidth (-3 dB) and passband ripple shall be measured using a constant amplitude unmodulated RF input signal which produces an output (S+N)/N ratio of 20 dB at the peak of the passband ripple. Passband ripple measurements shall be made in frequency increments not larger than 100 Hz.

4.5.5.2.2 Reciprocal mixing. Reciprocal mixing shall be measured with the receiver operating in the SSB mode. An unmodulated signal shall be applied at resonance to the input of the receiver from a 50-ohm standard dummy antenna (see 6.4.1) and adjusted to a level of -107 dBm available power at the receiver input. An unmodulated interfering signal shall be applied from the same standard dummy antenna at a frequency differing from the original signal by 100 kHz; the signal level shall be increased to the point where the (S+N)/N ratio of the original signal is decreased to 10 dB or the signal level reaches -17 dBm.

4.5.5.2.3 Blocking. Blocking shall be measured with the receiver in the SSB mode, 3 kHz bandwidth, and an unmodulated resonant wanted signal applied to the receiver input from a 50-ohm standard dummy antenna (see 6.4.1) and adjusted to yield a level of -87 dBm. The receiver gain shall be adjusted so that the wanted signal is just below the AGC threshold. An unmodulated unwanted signal shall be applied from the same standard dummy antenna at a frequency differing from the original signal by 100 kHz. The signal level shall be increased to the point where the (S+N)/N level of the original signal is decreased to 10 dB and the unwanted level shall be noted. In addition, the level at which the original output signal level is reduced by 3 dB shall be noted. The procedure shall be repeated for increasing frequency spacing of the unwanted signal from the receiver operating frequency by factors of two.

4.5.5.2.4 Cross mod. Cross mod shall be measured with the receiver in the SSB mode, 3 kHz bandwidth and an unmodulated signal at the resonant frequency applied to the receiver input from a 50-ohm standard dummy antenna (see 6.4.1) and adjusted to yield a level of -77 dBm. The receiver gain shall be adjusted so that the wanted signal is just below the AGC threshold. An unwanted

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signal amplitude modulated 30 percent by a 400 Hz tone shall be applied from the same dummy antenna at a frequency differing from the original signal by 100 kHz. The signal level shall be increased to the point where the receiver wanted output is modulated to a depth of 10 percent (cross mod sidebands less than 26 dB) or the input level reaches +12 dBm. The procedure shall be repeated for increasing frequency spacing of the unwanted signal from the receiver operating frequency by factors of two.

4.5.5.3 Spurious responses. Each primary frequency band shall be completely explored for evidence of the types of spurious responses specified in a through e which fall outside of the allowable limits specified in 3.13.3:

a. Internally generated spurious outputs (AGC off).
b. Image rejection. Measurements shall be made at the nominal end frequencies of each primary band, for all mixer images.
c. IF rejection. Measurements shall be made at the nominal end frequencies of each primary band.
d. Other single-signal spurious responses.
e. Front-end intermodulation shall be measured with the receiver operating in the SSB mode. Unmodulated signal pairs shall be used at frequencies specified in 3.13.3.5. Signal pairs shall be applied from a 50-ohm linear combiner and their level increased until the audio output reaches 10 dB (S+N)/N or the specified input level for each signal. Sufficient filtering or isolation shall be used between each generator and the combiner to ensure adequate freedom from intermodulation in the test signal.

4.5.5.3.1 Internal spurious responses. Internally generated spurious responses shall be measured in USB and LSB modes for receiver operation from both internal and external frequency standards. Initial measurements shall be made in the USB mode with receiver operation from the internal frequency standard. The receiver antenna input terminals shall be terminated with 50 ohms, AGC turned off. The receiver shall be tuned to 2.0990 MHz and the USB output signal monitored with an audio spectrum analyzer and earphones. The receiver shall be tuned throughout the complete frequency range of 14 kHz to 30 MHz in increments of not more than 100 kHz with dial settings of xx.xx90 MHz. Spurious responses detected shall be evaluated against the requirements of 3.13.3.1 by substitution of a standard signal generator for the 50-ohm input termination, offsetting the generator frequency to permit amplitude comparision of the internal spurious response and that due to the applied signal, using the audio spectrum analyzer. Measurements shall be repeated for LSB operation, using dial settings of xx.xx10 MHz.

4.5.5.4 Audio output measurements. Audio output measurements shall be as specified in 4.5.5.4.1 through 4.5.5.4.5.

4.5.5.4.1 Audio output voltage regulation. Measurements of audio output variation with change of load impedance and with variation of input signal level shall be made at the line output of the audio amplifier in USB and LSB. Tests shall be performed to ensure that no damage to output transistors occurs when operated into short and open circuit loads under zero drive and maximum drive conditions to verify compliance with 3.13.4.1.

4.5.5.4.2 Maximum undistorted output. Measurements shall be made in the AM, USB, and LSB modes at a nominal RF input frequency of 2.1 MHz. The AGC shall be operative, and measurements shall be made for all positions of the AGC time constant selector switch. Measurements shall be made on both rear-mounted line output receptacles and panel-mounted head telephone jacks. AF output data shall be measured for applied RF signal inputs from 1000 µV up to 1 V, with input level set in decade steps. Measurements shall be made at audio frequencies of 300 Hz, 1000 Hz, and 3500 Hz.

4.5.5.4.3 Range and linearity of audio level controls. Measurements shall be made of line output and head telephone control characteristics in the USB mode. AGC shall be active in the SLOW position of the AGC switch. Audio output as a function of angular rotation of the audio gain controls in increments of 10 degrees or less shall be measured starting from the full counterclockwise position. The measurements shall be performed at a nominal operating frequency of 2.1000 MHz.

4.5.5.4.4 Audio level meter. Measurements shall be made to verify compliance with 3.13.4.3.1.

4.5.5.4.5 Output hum and noise. Measurements shall be made for AM, USB, and LSB modes at RF input frequencies of 2.1 MHz, 17.1 MHz, and 29.9 MHz. The input signal level shall be 1000 µV through the antenna connector. For SSB modes, the signal shall be CW; for AM, the signal shall be modulated 90 percent at 1000 Hz. Measurements shall be made for power line frequencies of 60 Hz and 400 Hz, as specified in a and b:

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a. Set receiver RF gain to maximum and adjust line level control to produce standard output of 1 mW. With this level as a reference, measure relative amplitude of power line related components in the output, using a wave analyzer.

b. Set receiver line level control to maximum gain, and adjust the manual RF gain control to produce standard output of 1 mW. With this level as a reference, remove the RF input signal and measure the relative amplitude of the remaining hum and noise components with a broadband voltmeter.

4.5.5.5 Manual RF gain control. Range and linearity measurements shall be made with the receiver conditioned for USB (AGC inoperative) for an applied nominal RF input frequency of 2.1 MHz. Gain reduction in dB as a function of angular rotation in intervals of not greater than 10 degrees shall be measured.

4.5.5.6 AGC characteristics. Receiver AGC amplitude characteristics and time constants shall be measured for USB, LSB, and AM operation at the receiver low gain frequency. AGC linearity shall be verified. Measurements shall be made of maximum change in audio output level for RF inputs between that which produces standard output (S+N)/N (see 6.4.2) in the SSB modes and 1 V, open circuit.

4.5.5.6.1 AGC time constants. Separate determination of each of the AGC time constants shall be made by switching the RF input amplitude between the level which produces standard output response and various large signal levels from 100 µV to 0.1 V, open circuit, and observing the audio output. Oscilloscope photographs of the receiver audio output shall be obtained for attack and release characteristics, with oscilloscope sweep triggered on the RF amplitude transition.

4.5.5.6.2 Resonant overload. Measurement shall be made in the USB, LSB, and AM (90 percent mod) modes at the receiver high gain frequency. The receiver AGC shall be disabled, and AF and RF level controls shall be set for maximum gain.

4.5.5.6.3 Time of recovery from surge inputs. Measurements shall be made in USB and LSB modes at the receiver high gain frequency.

4.5.5.7 BFO control. The output frequency range of the BFO control shall be measured at an RF input frequency of 2.1 MHz.

4.5.5.8 In-band intermodulation. Measurements shall be made at receiver high and low gain frequencies, in USB and LSB modes, and for FAST and SLOW AGC modes. RF signal frequencies shall be made with RF signal amplitudes of -37 dBm per tone and -13 dBm per tone, and audio output of 3 Vrms per tone.

4.5.5.9 RF input circuit. Threshold of operation of the antenna input protection circuit and insertion loss in the operational condition shall be measured at the mid-frequency in each primary (MHz) band to verify compliance with the requirements of 3.13.9. Receiver input circuit protection shall be verified by keying drive to a suitable RF power amplifier held in a keydown condition. The power amplifier shall provide not less than 70 V open circuit in series with 50 ohms; the protection circuit shall be operated a minimum of 10 times at each test frequency. Circuit operating threshold shall be measured at 100 kHz, 2.5 MHz, 4.5 MHz, 8.5 MHz, 16.5 MHz, and 29.9999 MHz.

4.5.5.10 Accuracy of main tuning indicator calibration. Measurement shall be made at frequencies required to verify proper operation of each frequency selector in each position to verify compliance with the requirements of 3.3.2 and 3.13.10.

4.5.5.10.1 Tuning time measurement. Measurement shall be made of the maximum switching times for each frequency selector.

4.5.5.11 Frequency and gain stability. Frequency and gain stability shall be measured in accordance with the conditions specified in 4.5.5.11.1 through 4.5.5.11.6.

4.5.5.11.1 Sixty-minute warm-up period. The warm-up period shall begin after exposure of the receiver to the ambient temperature for not less than 6 hours with no power applied, to establish the temperature at the same value as the ambient temperature. Frequency measurements shall be made at intervals of not more than 2 minutes for the first 12 minutes, with the first measurements made 2 minutes after power is applied. Measurements shall be made at intervals of not more than 5 minutes for the next 18 minutes, and at intervals of not more than 10 minutes for the last 30 minutes.

4.5.5.11.2 Line voltage and frequency variation. Measurements shall be in accordance with the Steady state voltage and frequency paragraph of MIL-E-16400. A minimum stabilization period of 30 minutes shall be observed following each change of supply voltage and frequency.

4.5.5.11.3 Twenty-four-hour frequency drift. All receiver power shall be removed for at least 6 hours prior to stability measurements. The receiver shall be energized for a stabilization period not to exceed 48 hours, and operated during the test period with temperature, humidity, and line voltage controlled as specified in 3.13.11.1.6.

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4.5.5.11.4 Twenty-four-hour gain stability. Gain stability shall be measured in the ISB mode (signal in both sidebands), with AGC off. Receiver conditioning periods and temperature, humidity, and line voltage control, shall be as specified in 4.5.5.11.3.

4.5.5.11.5 Control variation. Measurement shall be made to verify compliance with 3.13.11.1.5.

4.5.5.11.6 Frequency standard adjustment range. Measurements shall be made to verify compliance with 3.13.14.

4.5.5.12 Regeneration. The complete tuning range shall be investigated for evidence of regeneration with the receiver operating at the upper steady-state line voltage limit.

4.5.5.13 Oscillator radiation. Measurements shall be made at the nominal end frequencies of each primary band.

4.5.5.14 Frequency standard calibration. Measurements shall be made to determine compliance with the requirements of 3.13.14 through 3.13.14.3.

4.5.5.14.1 External frequency standard. Operation using an external frequency standard shall be made to verify compliance with 3.13.14.1.

4.5.5.15 Frequency standard switch functions. Frequency standard assemblies shall be inspected to verify compliance with 3.13.14.3. This may be a qualitative test in accordance with 4.5.4.

4.5.5.16 Temperature and humidity. Tests shall be conducted in accordance with the Environmental tests paragraph for Range 4 of MIL-E-16400. Measurements of receiver gain variations (AGC off) and frequency stability shall be made to determine compliance with the requirements of 3.13.11.1.2 and 3.13.11.1.4. Measurements shall be made in USB, LSB, and CW modes at receiver high and low gain frequencies. Frequency accuracy (synthesizer lock) checks shall be made during temperature operating tests at the end of the period of operation at 0°C (32°F) after 4 hours of operation with the receiver stabilized at +50°C (122°F) and after return to room temperature. During the last hour of the high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the receiver under test shall be controlled to 50°C (122°F) by radiation-baffled heating and cooling sources as required (see 4.5.5.16.1). During humidity tests, measurements shall be made at the end of the second and fifth cycles (see 3.13.10 and 4.5.5.10).

4.5.5.16.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient temperature is visible to the unit under test. The maximum allowable temperature difference between any two points in the chamber shall be 10°C (18°F). The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant within ±1°C (2°F) of any selected test temperature.

4.5.5.16.2 Special requirements for humidity tests. During the checks performed at the end of the second and fifth cycle of the humidity test, the receiver shall be energized for a maximum of 30 minutes to verify proper operation. At the end of the 30 minutes, frequency for SSB modes shall be within two parts in 10^7 of the nominal frequency and maximum permissible gain change (AGC off) shall not cause audio output levels to vary by more than 4 dB (total) from those recorded as references at the end of the initial dry-out period.

4.5.5.17 Shock and vibration. Shock and vibration tests shall be performed as specified in MIL-S-901 and MIL-STD-167-1 and as specified in a and b:

a. Prior to and after completion of these tests, the tests specified in 1 through 3 shall be performed:

1. Sensitivity measurements in accordance with 4.5.5.1 in the CW, USB, and LSB modes for the 14 kHz, 200 kHz, 2 MHz, and 29 MHz bands only.
2. Synthesizer lock checks resulting from 4.5.5.10 testing.
3. Intermodulation distortion in accordance 4.5.5.8. Measurements of gain and frequency stability shall be made prior to and after testing. Measurements shall be made as specified in 3.13.11.

b. During vibration tests, the receiver shall be operated in the USB mode with the AGC time constant selector in the SLOW position, and with an unmodulated RF input signal producing an audio output of 1000 Hz to 3000 Hz. The RF input signal shall be continuously monitored with an oscilloscope and a wave analyzer. Any vibration-related product present at the audio output shall be not less than 30 dB below the reference audio output amplitude. No signal loss (dropout) exceeding 3 dB may occur during the vibration test.

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4.5.5.18 Supply line voltage and frequency variation and transients. Tests shall be performed in accordance with the Electrical tests paragraph of MIL-E-16400. Reference gain, sensitivity, and frequency measurements shall be made prior to start of testing. Variation of frequency and gain with line voltage and frequency shall be measured against the requirements of 3.13.11.1 for a nominal line voltage of 115 VAC and for a nominal line frequency of 60 Hz. Following performance of the electrical tests specified in MIL-E-16400, no permanent changes shall be produced in frequency, gain, or sensitivity.

4.5.5.19 Power interruption and spike tests. Power interruption tests shall be performed in accordance with the Power interruption paragraph of MIL-E-16400. The 2500 V spike test shall be performed in accordance with DoD-STD-1399, Section 300. Source impedance for the spike shall be 25 ohms, resistive. Spikes shall be delivered line-line and line-ground for each primary power line. For each test, a positive-going spike shall be applied at the positive crest of the supply line voltage waveform and at the zero axis crossing, and a negative-going spike shall be applied at the negative crest.

4.5.5.20 Power and power factor. Tests shall be conducted in accordance with the Power and power factor paragraph of MIL-E-16400.

4.5.5.21 Chassis orientation. Verification of the ability to orient the chassis ±90 degrees shall be performed.

4.5.5.22 Assembly identification. All applicable assemblies shall be inspected to verify compliance with 3.12.1.13.

4.5.5.23 EMC. Compliance with 3.13.16 shall be verified by tests performed in accordance with MIL-STD-462. Group C inspection shall be performed for tests CE01, CE03, CS06, and RE02 and any other tests requested by the Navy for this procurement. Tests shall be performed at a power line frequency of 60 Hz.

4.5.5.23.1 EMP. The requirements of 3.13.16.1 shall be verified as specified in a and b:

a. The requirements of 3.13.16.1a shall be verified by an analysis or test of a pulse induced from a simulated exo-atmospheric nuclear explosion whose RF field is of the following form and amplitude:

$$E(t) = E_o(e^{-\alpha t} - e^{-\beta t}) \text{ and}$$

$$H(t) = \frac{E(t)}{Z_o}, \text{ where}$$

E(t) = Electric field in volts per meter (V/m) at time, t
H(t) = Magnetic field in A per meter at time, t
E_o = 5.25×10^4 V/m
α = 4.0×10^6 per second
β = 4.76×10^8 per second
Z_o = 377 ohms
t = Time in seconds

b. The requirements of 3.13.16.1b shall be verified by the test method specified in 1 through 6:

1. Applicable test frequencies. Tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz, 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz, and 100 MHz. In addition, tests shall be performed at the test sample's critical frequencies, such as local oscillator, power switching frequencies and harmonics, RF frequencies, clock frequencies, bandpass frequencies, and so forth, as specified in the EMI test plan.

2. Apparatus. Test apparatus shall consist of the items specified in A through D:

A. Source injection pulse generator with the capabilities specified in (1) through (4):

(1) Damped sinusoidal transient as shown in the applicable limit and as described by the equations in (a) and (b):

(a) $I_{pin}(t) = 1.05\ I(f)\ e^{-0.2ft}\ SIN\ (2\pi ft)$

(b) $V_{pin}(t) = 1.05\ V(f)\ e^{-0.2ft}\ SIN\ (2\pi ft)$

where
$I_{pin}(t)$ = Pin to case current, in A at time, t
$V_{pin}(t)$ = Pin to case voltage, in V at time, t
f = Specific test frequency, in MHz
t = Time in seconds
I(f) = Current determined from FIGURE 2
V(f) = Voltage determined from FIGURE 3

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(2) Variable amplitude so the injected transient can be increased from a minimum to a maximum specified level (that is, at least 10 percent, 50 percent, and 100 percent of the amplitude must be tested at each specified test frequency)
(3) Frequency range capable of generating the required test frequencies
(4) Current output level of at least 10 A into a 100-ohm load
B. Current probe
C. Oscilloscope with a minimum bandwidth capability of 100 MHz
D. Interconnecting test cable

3. Test setup and procedure. The test setup and procedure shall be as specified in 4 and 5. A functional analysis shall be performed before the certification testing to identify the items specified in A through C:
A. The interface pin connectors
B. The test sample functions which must be monitored to determine system failure or transient upset
C. The additional test frequencies (that is, interface cable resonant frequency, clock frequencies, data rates, and so forth)

4. Test setup. Test setup shall be as shown in FIGURE 4.

5. Test procedure. The test procedure shall be as specified in A through E:
A. Apply the test signal to each interface pin of the test sample at each test frequency
B. Monitor the test sample functions in all modes of operation to determine whether there is a failure or transient upset
C. Inject one transient pulse on each interface pin and at each test frequency
D. The test sample shall be tested with its power off and on
E. Record test frequency and level at which a failure or upset occurs

6. Notes:
A. The test sample shall be tested on a ground plane. There shall be no insulated materials between the test setup and the ground plane.
B. The test cable running between the test sample and the damped sinusoidal generator shall be kept 4 cm to 6 cm above the ground plane.

4.5.5.24 Remote operation. All remote controlled functions shall be verified in accordance with 3.13.17 and 4.5.4.

4.5.5.25 COR. The COR shall be verified to comply with 3.13.18. This may be a qualitative test in accordance with 4.5.4.

4.5.5.26 Leakage current. Leakage current shall be measured at 115 V, 60 Hz. Leakage current shall be measured on one unit of the first month's production. Thereafter, leakage current shall be measured on one unit for each successive 100 units produced.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE RECEIVER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.5.26.1 Receiver connections. Each receiver directly connected to an external power source and units deriving power from the receiver shall be placed on an insulated surface. All safety ground conductors between the receiver and units deriving power from the receiver shall be intact. The safety ground conductor between the receiver under test and the source power shall be opened during the test. OBSERVE WARNING STATEMENT. The receiver shall be connected as shown in FIGURE 5 if it is connected to single-phase source power.

4.5.5.26.2 Measurement. Leakage current shall be measured on the receiver in its normal operating configuration. Receiver controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage and the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the receiver is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 5. The open safety ground conductor shall be reconnected immediately after the test is completed.

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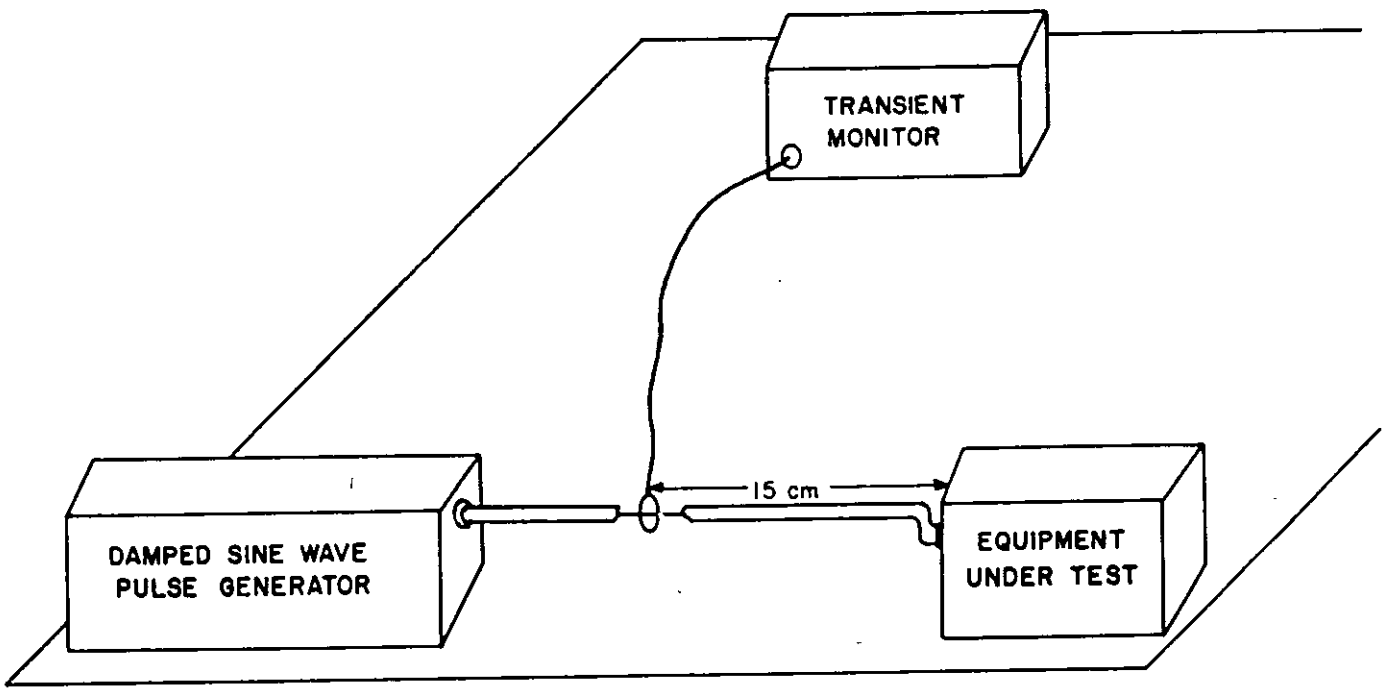
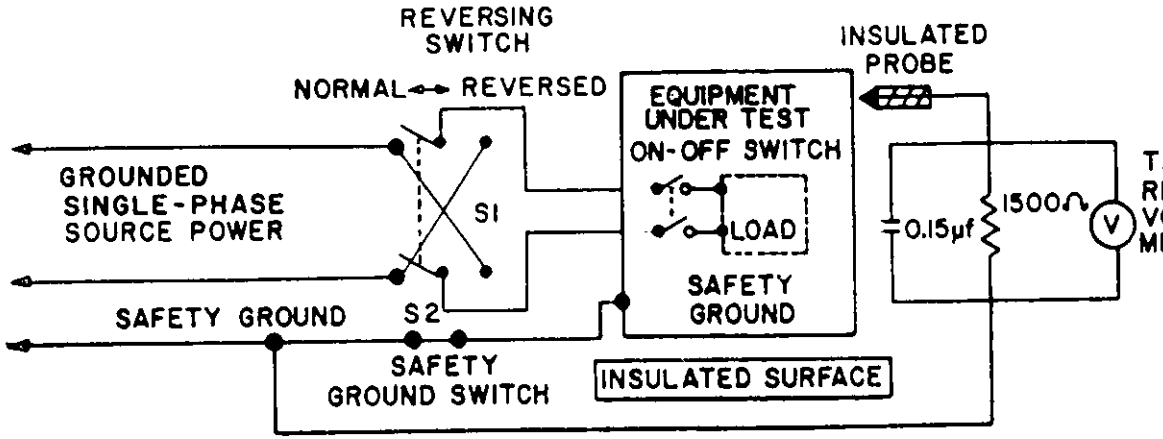


FIGURE 4. EMP test setup.

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WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

GENERAL ORDER OF TEST:

1. Source power OFF. Connect equipment per diagram.
2. ON-OFF SW OFF. S1 SW normal. S2 SW closed. Connect source power.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. For each probe point, record voltmeter reading (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. Repeat Step 4.
6. S1 SW REVERSED. ON-OFF SW ON. Repeat Step 4.
7. ON-OFF SW OFF. Repeat Step 4.
8. S2 SW CLOSED. S1 SW normal.
9. Repeat Step 3 THRU 8 for each mode of operation.
10. Remove source power. Disconnect equipment. Sign record sheet.

FIGURE 5. Single-phase test diagram for leakage current measurement.

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4.5.6 Test by analysis. When 4.5.6 is specified in TABLE VI, the test may be performed by analysis or calculation, or both.

4.6 Standard performance test conditions. Standard performance test conditions shall be as specified in 4.6.1 through 4.6.8.

4.6.1 Primary power. The nominal primary power test condition shall be 115 VAC, 60 Hz, single-phase.

4.6.2 Ambient temperature. Except where otherwise specified herein, the nominal test environment temperature shall be 25°C (77°F).

4.6.3 Relative humidity. The relative humidity shall be less than, or equal to, 95 percent.

4.6.4 External - internal frequency standard switch. The frequency standard (FREQ STD) switch shall be set at its INT position.

4.6.5 AF output. Unless otherwise specified herein, all AF outputs shall be measured with a 600-ohm nonreactive load connected across the output terminals.

4.6.6 RF input to receiver. All RF inputs to the receiver shall be applied to the antenna input terminals through a standard dummy antenna.

4.6.7 Receiver BFO control. Normally, the BFO control shall be adjusted at the zero or mid-scale position, but may otherwise be adjusted, as required in 3.13.7, to determine the performance of the receiver.

4.6.8 Local and remote operation. The local and remote switch shall be in the local mode.

4.7 Reliability qualification test. A reliability qualification test shall be performed in accordance with Test Plan IVC of MIL-STD-781. At least three units from the first month's production shall comprise the test lot. Environmental conditions, operating cycles, and performance shall be as specified herein.

4.7.1 Test failure (see 6.4.15). In addition to the failure definition of 6.4.15, any parameter deviation which requires adjustment or alignment of the receiver under test shall be classified as a failure. Receiver failures occurring during the test shall be handled in accordance with the Failure actions paragraph of MIL-STD-781.

4.7.2 Environmental conditions for reliability testing. The environmental conditions specified in a through d shall be maintained throughout the reliability acceptance testing:

a. Temperature in the test chamber shall be cycled between 0°C (32°F) ±2°C (4°F) and 50°C (122°F) ±2°C (4°F) in accordance with FIGURE 6. The temperature rate of change shall be not less than 5°C (9°F) per minute.

b. A 4 Hz to 33 Hz sine sweep vibration shall be applied continuously in a single axis at the amplitude of 0.51 mm (0.020 in.) (double amplitude (DA)) ±0.10 mm (0.004 in.) DA. The sweep time shall be 10 minutes (up and down).

c. Input voltage shall be 115 VAC, 60 Hz. Input voltage shall be cycled in accordance with the 3-day cycle specified in 1 through 3:

1. Nominal
2. Nominal +10 percent
3. Nominal -10 percent

d. Relative humidity: Ambient

4.7.3 Performance tests. At least daily performance measurements shall be made in accordance with the tests specified in a and b. The receiver shall be set to the internal frequency standard.

a. Gain and sensitivity: Receiver sensitivity and gain shall conform to the requirements of 3.13.1.1 and 3.13.1.2, respectively, for all six modes of operation specified in TABLE VII.

b. Frequency stability: The drift of the internal frequency standard shall be not greater than 5 parts in 10^8 per day, nor 3 parts in 10^7 per month.

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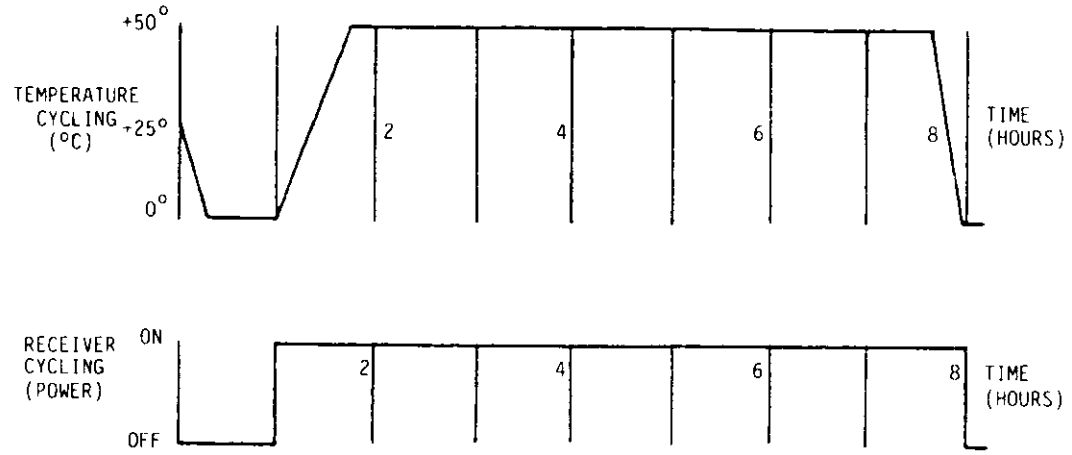


FIGURE 6. Reliability temperature and receiver cycling.

4.7.4 Reject decision. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall:

a. Develop an approach for redesign or correction of all deficiencies.
b. Using the selected approach, repeat the reliability qualification test until an accept decision is reached.
c. Implement a program of corrective action for all units previously delivered by the contractor under the current contract.

TABLE VII. Test modes and frequencies.

Channel	Mode	Test frequency (MHz)	Bandwidth (kHz)
1	AM	2.0015, 0.2000	6
2	FM	3.2750, 0.3000	16
3	CW	5.3340, 0.0140, 0.0500	0.3
4	ISB	9.5000	3
5	USB	16.0015	3
6	LSB	29.9909	3

4.8 Maintainability demonstration. The maintainability demonstration shall be as specified in 4.8.1 through 4.8.8.

4.8.1 First article inspection maintainability demonstration. As a part of the first article inspection, a maintainability demonstration shall be performed at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the receiver MTTR requirement is fulfilled.

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4.8.1.1 Demonstration conditions. The receiver maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Fifty candidate corrective maintenance faults shall be determined in accordance with Appendix A of MIL-STD-471. The receiver information specified in a through c shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity, or its authorized representative, shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.8.2 Accept or reject criteria. Accept or reject criteria shall be as specified in 4.8.2.1.

4.8.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE VIII. Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index, MTTR or M_{maxct} is less than, or equal to, that shown in TABLE VIII corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and determined as greater than or less than each index. The actual value shall also be determined. An accept decision can be made only when an accept decision is made for both MTTR and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test shall have resulted in a reject decision. More than one erroneous failure indication shall constitute failure of the demonstration.

TABLE VIII. Acceptance criteria.

Criterion	MTTR	M_{maxct}
Acceptance level	8	0
Sample size	20	20

4.8.3 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the receiver shall not interfere with on-line operation.

4.8.4 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the receiver will be installed.

4.8.5 Technicians. The Government reserves the right to provide and select technicians to perform the maintainability demonstration.

4.8.6 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the receiver.

4.8.7 Rejection of demonstration. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall:

a. Develop an approach for redesign or correction of all deficiencies.
b. Upon approval of an approach, repeat the demonstration until an accept decision is reached.

4.8.8 Acceptance criteria. The maintainability demonstration shall demonstrate that the receiver conforms to the acceptance criteria prior to delivery.

4.9 Workmanship. Workmanship shall be as specified in 4.9.1 through 4.9.2.1.

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4.9.1 General workmanship. The receiver, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process. Each solder connection and associated wiring leads shall be visually examined.

4.9.2 Workmanship screen. Workmanship screen shall be as specified in 4.9.2.1.

4.9.2.1 Temperature cycling. Each receiver shall be subjected to 10 cycles of the temperature curve shown in FIGURE 7. The temperature rate of change shall be not less than 5°C (9°F) per minute. Receiver power shall be turned on and off at the indicated times. The receiver shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the receiver reaches 80 percent of the chamber temperature. When failures occur, the receiver shall be reworked and the cycling continued for a cumulative total of 10 cycles. Receiver operating time of 20 hours shall be accumulated during these 10 cycles.

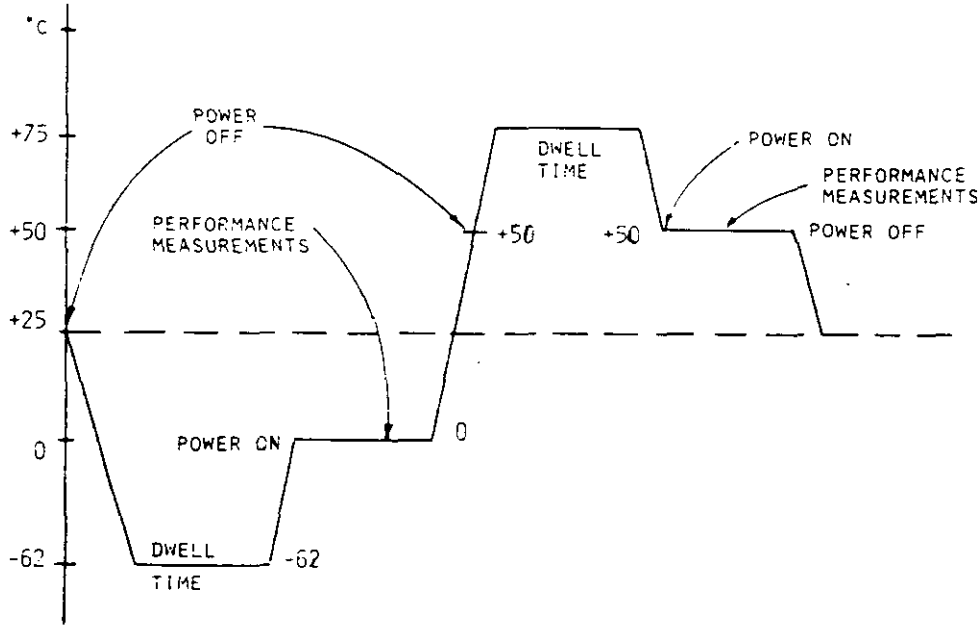


FIGURE 7. Temperature curve (one cycle) (not to scale).

4.10 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The receiver covered by this specification is intended for general purpose voice, CW, FSK, AM, LSB, and USB FM and may be used for surface ship, submarine, and shore installations.

6.1.1 Mission profile. The mission profile for the receiver is continuous operation while underway. A 15-year life is planned unless the prediction and tests indicate otherwise.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected production control and environmental test units may not be accepted on the contract (see 4.4.4)
d. Levels of preservation, packaging, packing, and marking (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of the Defense Acquisition Regulations 7-104.9(n)(2) are invoked and the DD form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.3, 4.4	First Article and Quality Conformance Inspection Test Procedures	UDI-T-22710B
4.3, 4.4	First Article and Quality Conformance Inspection Test Report	DI-T-5247
4.7	Reliability Test and Demonstration Reports	DI-R-7034
4.7	Reliability Test Procedures	DI-R-7035/M
4.7.1	Corrective Action Plan	DI-R-7038
4.8	Maintainability Demonstration Test Plan	DI-R-7112

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulation. The first article should be a first production item. The first article should consist of three units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, tests, and approval of the first article.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.15.

6.4.1 Standard dummy antenna. The standard dummy antenna is a device whose electrical constants simulate the impedance characteristics of antennas suitable for use with each receiver throughout its frequency range.

6.4.1.1 Dummy antenna characteristics. A device conforming to the requirements of 6.4.1 is a 50-ohm nonreactive resistor. This resistance includes the output resistance of the standard signal generator. If necess

6.4.1.2 Application and intent. The intent of 6.4.1.1 is that the standard signal generator produce the equivalent of the signal voltage induced by an electromagnetic field in an antenna which has the constants that are simulated by the dummy antenna. Hence, the voltage reading obtained from the standard signal generator is the open circuit voltage in series with the dummy antenna. This definition applies to all voltages specified relative to the receiver antenna input terminals. Many standard signal generators do not indicate open circuit voltage, but rather the voltage which appears across a matched load. For these generators, the desired open circuit voltage is equal to twice the indicated output.

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6.4.2 Standard output (S+N)/N ratio. Standard output (S+N)/N is 10 dB. This ratio should be determined in terms of rms values.

6.4.2.1 CW and sideband reception modes. For CW and sideband reception modes, the output (S+N)/N ratio is the ratio between the audio output resulting from an unmodulated RF input signal producing the standard beat note frequency and the output due to noise when the RF input signal is removed.

6.4.2.2 AM reception. For AM reception, the output (S+N)/N ratio is the ratio between the audio output resulting from the RF input signal modulated 50 percent at the standard mod frequency and the output due to noise when the mod is removed from the RF carrier.

6.4.3 Standard audio output. Standard audio output is a nominal 1 mW dissipated into a 600-ohm nonreactive resistance across the AF line output terminals.

6.4.4 Standard gain. Standard gain is that gain which produces standard output (S+N)/N ratio (see 6.4.2).

6.4.5 Standard mod frequency. Standard mod frequency is 1000 Hz.

6.4.6 Standard beat note frequency. Standard beat note frequency for CW and SSB reception is 1000 Hz.

6.4.7 Sensitivity. Sensitivity of the receiver is that characteristic which determines the minimum strength of an input signal capable of producing a desired output (S+N)/N ratio.

6.4.7.1 Standard CW sensitivity. Standard CW sensitivity is the number of μV (open circuit) of unmodulated signal from a 50-ohm standard dummy antenna required to produce a standard output (S+N)/N ratio (see 6.4.2.1).

6.4.7.2 Standard AM sensitivity. Standard AM sensitivity is the number of μV of signal from a 50-ohm standard dummy antenna, modulated 50 percent, required to produce a standard output (S+N)/N ratio (see 6.4.2.2).

6.4.8 Selectivity. Selectivity of the receiver is that characteristic which determines the extent to which it is capable of discriminating between a desired RF signal and any undesired signals at frequencies required by the desired signal information spectrum but outside the bandwidth.

6.4.8.1 Antenna input to detector input. Receiver RF and IF selectivity is determined primarily by the degree to which the information band filters are capable of rejecting undesired signals at frequencies immediately adjacent to the desired signal information spectrum. Rejection of undesired signals is expressed in terms of dB with respect to the peak of the desired signal passband ripple.

6.4.8.2 Overall passband (antenna input and audio output). Selectivity is determined by the combined effects of all frequency selective circuits in the entire signal path.

6.4.9 Spurious responses. In general, a spurious response in a receiver is the presence of any signal in the bandpass of the IF amplifiers which is not a function of the desired signal alone. There are several specific classes of spurious response as specified in a through d:

a. Internally generated spurious outputs caused solely by the presence of the oscillators inherent to the receiver
b. Signals which require the presence of a single strong signal external to the receiver; specific types are IF and image responses
c. Signals resulting from the intermodulation of two strong external signals in the first stages of the receiver prior to the IF amplifier
d. The superimposition of mod from a single strong signal external to the receiver on a desired signal by a cross mod process.

6.4.9.1 Image rejection. Images are the undesired second order products of each conversion resulting from a signal on the opposite side of the injection frequency from the desired response. Image rejection is expressed as the ratio of input signal amplitudes, in dB at the image and resonant frequencies, for standard (S+N)/N ratio.

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6.4.9.2 IF rejection. IF rejection is defined as the degree to which the receiver is capable of rejecting signals having the frequency(ies) of the IF amplifier(s). IF rejection is expressed as the ratio of input signal amplitudes, in dB, at the IF and resonant frequencies, for standard output (S+N)/N ratio.

6.4.10 Resonant overload. Resonant overload output is the output at which the overall gain (with no AGC) of the receiver as conditioned for standard CW sensitivity, falls 3 dB with an increase of input carrier level.

6.4.11 Maximum undistorted output. Maximum undistorted power output is the output power level at which the total output harmonic content is 5 percent, in voltage, for an input carrier adjusted for the standard sideband measurement frequency (see 6.4.12) at the level of 1000 µV.

6.4.11.1 Total harmonic content. The total harmonic content is determined by measuring the residual voltages with the distortion analyzer after eliminating the fundamental of a sinusoidal modulating frequency.

6.4.12 Standard sideband measurement frequency. Standard sideband measurement frequency is 1000 Hz higher than the carrier for the USB and 1000 Hz lower than the carrier for the LSB.

6.4.13 AGC time constants. AGC time constants are as specified in a through c:

a. Attack time: The time from application of the large RF signal level until the audio output amplitude has stabilized within 3 dB of the final steady-state output
b. Decay time: The time after removal of the large RF signal, for the audio output voltage to rise from 10 percent to 63 percent of the final value due to the small signal
c. Hold (hang) time: The time from removal of the large RF signal level until the audio output voltage has risen to 10 percent of the steady-state output due to the small signal

6.4.14 AGC threshold level. AGC threshold level is the RF input voltage for which the RF input to AF output characteristic departs from linearity by 1 dB.

6.4.15 Test failure. Failure is defined as any malfunction or parameter deviation that prevents the receiver from performing within the specified operational requirements.

6.5 Maintenance concept. The receiver is part of a family of HF communications equipments which are designed to minimize equipment down time through the use of plug-in assemblies. Combined with the maintenance complexity resulting from high density packaging, this results in a maintenance program entailing organizational, intermediate, and depot level maintenance.

6.6 Environmental inspection. Approval to ship may be withheld at the discretion of the the Government pending the decision of the contracting officer on the adequacy of corrective action taken by the contractor.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
NAVY-EC

(Project No. 5820-N800(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-28886A	Receiver, Low Frequency, Medium Frequency and High Frequency Co

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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